29/05/2026

Code-P

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720 **Project Phoenix for Re-NEET-2026_CST-02P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. Two particles *A* and *B* are initially at rest at the origin. They are having acceleration $\vec{a}_A = \left(2t\,\hat{i}\right)$ m/s^2 and $\vec{a}_B = \left(4\,\hat{j}\right)$ m/s^2 respectively. The magnitude of relative velocity of *A* w.r.t. *B* at *t* = 2 s is

(1) $4\sqrt{5}$ m/s (2) 12 m/s
(3) 4 m/s (4) $2\sqrt{5}$ m/s

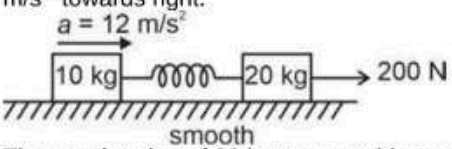
2. A bullet from a gun is fired on a wooden block with velocity u. When bullet travels 30 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{2}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is. [Assume that bullet undergoes uniform retardation]

(1) 37.5 cm (2) 42 cm
(3) 40 cm (4) 45 cm

3. A body starts moving with an acceleration of $0.5\ \text{m/s}^2$. Distance travelled by it in 7th half second is.

(1) 0.6250 m (2) 0.4850 m
(3) 0.8125 m (4) 1.425 m

4. Two masses 10 kg and 20 kg are connected by a light spring as shown in the figure. A horizontal force of 200 N acts on the 20 kg mass and the 10 kg mass has acceleration 12 m/s^2 towards right.



The acceleration of 20 kg mass at this moment will be

(1) Zero (2) 10 m/s^2
(3) 4 m/s^2 (4) 12 m/s^2

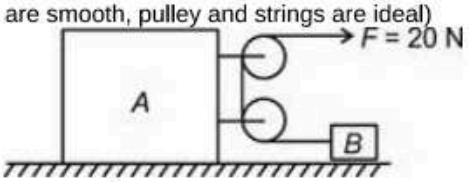
5. **Assertion(A):** For a rigid body, separation between any two points changes under the application of forces on these two points.
Reason(R): For a rigid body rotating about a fixed axis, the angular velocity has a variable direction.
In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

6. A particle of mass 300 g is moving with a speed of 20 ms^{-1} along the straight line $y = x - 4\sqrt{2}$. The angular momentum of the particle about the origin is (where *y* & *x* are in metres)

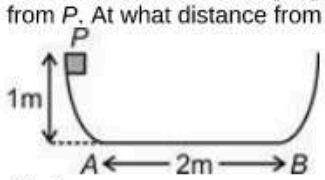
(1) 24 kg m^{2}s^{-1} (2) $24\sqrt{2}$ kg m^{2}s^{-1}
(3) 12 kg m^{2}s^{-1} (4) $6\sqrt{2}$ kg m^{2}s^{-1}

7. A cubical block *A* of mass 10 kg and a block *B* of mass 2 kg are connected via string-pulley system as shown in the figure. If one end of the string is pulled with a force 20 N as shown, then the acceleration of block *A* will be (All surfaces are smooth, pulley and strings are ideal)



(1) 4 m/s^2 (2) 2 m/s^2
(3) 1 m/s^2 (4) 8 m/s^2

8. The curved portion of the track shown, is smooth and horizontal surface is rough (μ = 0.2). The block is released from *P*. At what distance from *A* will it stop?



(1) 1 m (2) 2 m
(3) 3 m (4) 4 m

9. The change in the gravitational potential energy when a body of mass *m* is brought to surface of earth from a height *nR* above surface is (where *R* is the radius of earth)

(1) $\frac{-n}{n+1}mg$ (2) $\frac{-n}{n+1}mgR$
(3) $\frac{n}{n-1}mgR$ (4) $\frac{n+1}{n}mgR$

10. The terminal velocity of a metallic ball of radius 1 mm falling through a viscous liquid at 25°C is 2 cm/s. If density of ball is 4.2×10^3 kg/m^3, density of liquid is 1.2×10^3 kg/m^3, then viscosity of liquid at 25°C is nearly (g = 10 m/s^2)

(1) 3.3×10^{-1} kg m^{-1}s^{-1} (2) 1.5×10^{-2} kg m^{-1}s^{-1}
(3) 2×10^{-1} kg m^{-1}s^{-1} (4) 4×10^{-1} kg m^{-1}s^{-1}

11. In adiabatic change, the pressure and temperature of a linear triatomic gas are related as $P \propto T^k$, where *k* equals (ignore vibrational modes)

(1) $\frac{5}{2}$ (2) $\frac{5}{3}$
(3) $\frac{7}{2}$ (4) $\frac{7}{3}$

12. A cylinder of radius *R* made of a material of thermal conductivity *K* is surrounded by a cylindrical shell of inner radius *R* and outer radius 3*R* made of a material of thermal conductivity 2*K*. The two ends of the combined system are maintained at two different temperatures. There is no loss of heat across the cylindrical surface and the system is in steady state. The effective thermal conductivity of the system is

(1) $\frac{17K}{9}$ (2) $\frac{9k}{7}$
(3) $\frac{11K}{9}$ (4) $\frac{9K}{2}$

13. Consider the following statements.
Statement (A): Kirchoff's second law is based on conservation of energy.
Statement (B): Body moves under constant acceleration should move in a straight line.
Statement (C): Working of meter bridge is based on balance Wheatstone bridge.
Statement (D): Current is a scalar quantity while current density is a vector quantity.
Tick the correct option among the following.
(1) Statement (A), (B) and (C) are true
(2) Statement (A) and (C) are correct while Statement (B) and (D) are incorrect
(3) Statement (A), (C) and (D) are correct
(4) All statements are correct.

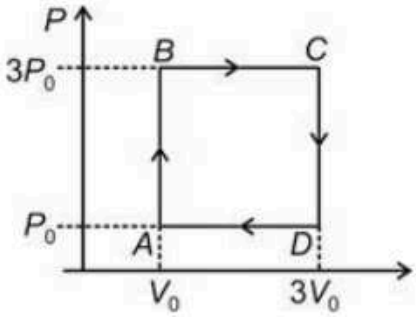
14. A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm in it. If the entire arrangement is put in elevator moving down with acceleration $\frac{g}{2}$, then height of water column in the capillary tube is

(1) 20 cm (2) 16 cm
(3) 8 cm (4) 4 cm

15. A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion 10^{-5} °C^{-1}, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

(1) 5×10^3 N (2) 50×10^3 N
(3) 100×10^3 N (4) 2×10^3 N

16. A fixed amount of an ideal monoatomic gas is taken through the cycle *ABCD* as shown in the figure. The efficiency of the cycle will be



(1) 22.2% (2) 33.33%
(3) 20% (4) 40%

17. 5 g of ice at –10°C is mixed with 20 g of water at 60°C. The equilibrium temperature of mixture will be (specific heat of ice $= 0.5\frac{cal}{g°C}$)

(1) 31°C (2) 25°C
(3) 0°C (4) 65°C

18. The frequency of tuning fork *A* and *B* are respectively 5% more and 3% less than the frequency of the tuning fork *C*. When *A* and *B* are simultaneously excited, 5 beats per second are produced, then frequency of fork *B* will be

(1) 70.625 Hz (2) 60.625 Hz
(3) 40.625 Hz (4) 50.625 Hz

19. A string has linear mass density 100 g/m and is under tension *T* = 90 N. A sinusoidal wave is sent with frequency $f = \frac{50}{\pi}$Hz and amplitude *A* = 1 cm along the string. At what average rate does the wave transport the energy?

(1) 1.5 mW (2) 1.5 W
(3) 3 mW (4) 3 W

20. An earthquake generates both transverse (*S*) and longitudinal (*P*) sound waves in the earth. The speed of *S* wave is about 6 km s^{-1} and that of *P* wave is about 12 km s^{-1}. A seismograph records *S* and *P* waves from an earthquake. The first *P* wave arrives 3 minutes before first *S* wave. The epicentre of earthquake located at the distance is about

(1) 1540 km (2) 1710 km
(3) 1918 km (4) 2160 km

21. Two bodies of masses M_1 and M_2 are separated by a certain distance. If $\vec{F}_{12}$ is the force on M_1 due to M_2 and $\vec{F}_{21}$ is the force on M_2 due to M_1 then which of the following relation is correct?

(1) $\vec{F}_{12} = \vec{F}_{21}$

(2) $\vec{F}_{12} + \vec{F}_{21} = 0$

(3) $\left|\vec{F}_{12}\right| = \left|\vec{F}_{21}\right|$

(4) Both (2) and (3)

22. Two moles of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ are mixed with four moles of monoatomic gas $\left(\gamma = \frac{5}{3}\right)$. The value of γ for the mixture is

(1) 1.2 (2) 1.8

(3) 1.54 (4) 1.67

23. A horizontal stretched string, fixed at two ends, is vibrating in its fifth harmonic, according to the equation $y(x, t) = (0.01\ \text{m})\sin((62.8\ \text{m}^{-1})x)\cos((628\ \text{s}^{-1})t)$.

Match column-I with column-II (where symbols have their usual meaning and assume 1st node is present at $x = 0$)

Column-I	Column-II (in S.I. units)
(A) Position of 2nd node	(P) 0.025
(B) Length of string	(Q) 0.05
(C) Position of 1st antinode	(R) 0.25

(1) (A)(P), (B)(Q), (C)(R) (2) (A)(Q), (B)(P), (C)(R)

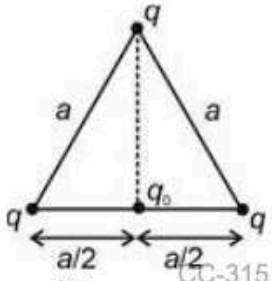
(3) (A)(P), (B)(R), (C)(Q) (4) (A)(Q), (B)(R), (C)(P)

24. Two equal resistances when connected in series to a battery, consume electric power of 100 W. If these resistances are now connected in parallel combination to the same battery, then electric power dissipated will be

(1) 800 W (2) 25 W

(3) 200 W (4) 400 W

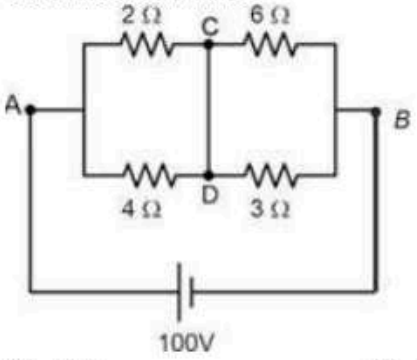
25. Each corner of an equilateral triangle of side a has charge q and a charge q_0 is placed at midpoint of side (shown in figure). Net force acting on charge q_0 will be $\left(K = \frac{1}{4\pi\varepsilon_0}\right)$



(1) $\frac{Kqq_0}{a^2}$ (2) $\frac{Kqq_0}{3a^2}$

(3) $\frac{4Kqq_0}{3a^2}$ (4) $\frac{3Kqq_0}{2a^2}$

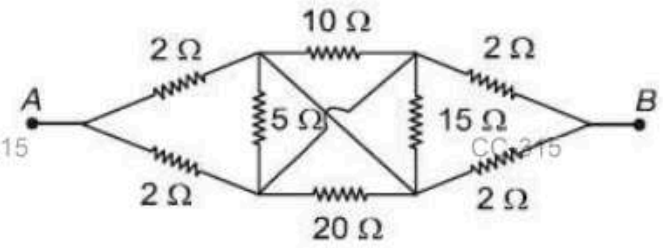
26. A battery of 100 V is connected between the points A and B of the circuit shown in the figure. Current through the branch CD of the circuit will be



(1) 10 A (2) 20 A

(3) 30 A (4) 7.5 A

27. The equivalent resistance between points A and B would be



(1) 5 Ω (2) 2 Ω

(3) 4 Ω (4) $\frac{15}{2}$ Ω

28. A bob of a simple pendulum has mass 2 g and a charge of 8 μC. It is at rest in a uniform horizontal electric field of intensity 1000 V/m at equilibrium. The angle that the pendulum makes with horizontal is

(1) $\tan^{-1}[5]$ (2) $\tan^{-1}[3]$

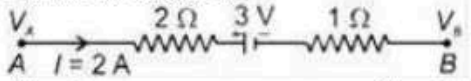
(3) $\tan^{-1}\left[\frac{3}{2}\right]$ (4) $\tan^{-1}\left[\frac{5}{2}\right]$

29. The minimum electrostatic force between two charged particles placed at a distance of 3 cm is

(1) 2.56×10^{-25} N (2) 1.28×10^{-25} N

(3) 2.56×10^{-30} N (4) 1.28×10^{-30} N

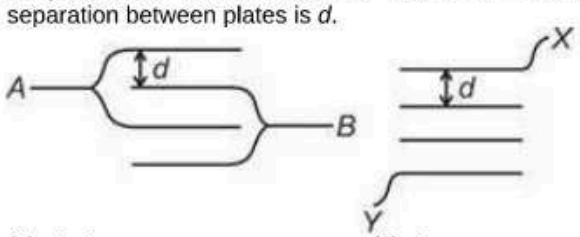
30. The potential difference $(V_A - V_B)$ between the points A and B in the given figure is



(1) –3 V (2) +3 V

(3) +6 V (4) +9 V

31. What is the ratio of effective capacitance between points A and B to the effective capacitance between points X and Y? All plates are identical, each of area A and successive separation between plates is d.



(1) 4 : 1 (2) 1 : 4

(3) 9 : 1 (4) 1 : 1

32. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1?



(1) 0, 1 (2) 0, 0
(3) 1, 0 (4) 1,1

33. In an electromagnetic wave in free space that root mean square value of the electric field is E_{rms}=6 V/m. The peak value of the magnetic field is

(1) $1.41 \times 10^{-8} T$ (2) $2.83 \times 10^{-8} T$
(3) $0.70 \times 10^{-8} T$ (4) $4.23 \times 10^{-8} T$

34. Two polaroids are placed such that their planes are parallel to each other with their axis of transmission at 30°. If an unpolarised light of intensity I_0 is incident on first polariser then intensity of light that will be transmitted through the second polaroid is

(1) $\frac{I_0}{2}$ (2) $\frac{I_0}{8}$
(3) $\frac{3I_0}{4}$ (4) $\frac{3I_0}{8}$

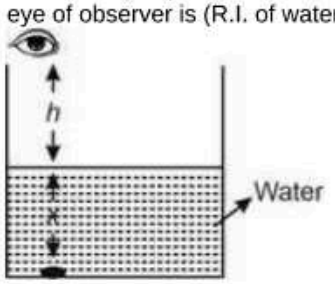
35. In Young's double slit experiment using sodium light of wavelength 5898 Å, 92 fringes are seen. If in same set up, light of wavelength 5480 Å is used, then number of fringes that can be seen are

(1) 62 (2) 67
(3) 85 (4) 99

36. The de Broglie wavelength of a neutron at 27°C is λ. What will be its wavelength at 327°C?

(1) $\frac{\lambda}{2}$ (2) $\frac{\lambda}{\sqrt{2}}$
(3) $\frac{\lambda}{3}$ (4) $\frac{\lambda}{2\sqrt{3}}$

37. A coin is seen from a height *h* above water surface (as shown in figure). The distance of the coin appearing from the eye of observer is (R.I. of water = μ)



(1) $h + \mu x$ (2) $h - \frac{x}{\mu}$
(3) $h + \frac{x}{\mu}$ (4) $x\left(1 - \frac{1}{\mu}\right) + h$

38. The minimum number of NAND gates required to form an AND gate

(1) 1 (2) 2
(3) 3 (4) 4

39. The focal length of a glass (μ = 1.5) lens in air is 20 cm. If it is dipped in water $\left(\mu = \frac{4}{3}\right)$ its focal length in water will be approximately equal to

(1) 80 cm (2) 40 cm
(3) 60 cm (4) 20 cm

40. A galvanometer has a resistance of 50 Ω and a full scale deflection current of 50 μA. It can be used as a voltmeter or as a higher range ammeter provided that a resistance is added to it. Choose the correct range and resistance combinations from the following

(1) 1 V range with approximately 200 kΩ resistance in series
(2) 100 V range with approximately 200 kΩ resistance in series
(3) 1 mA range with 50 Ω resistance in parallel
(4) 0.1 mA range with 50 Ω resistance in parallel

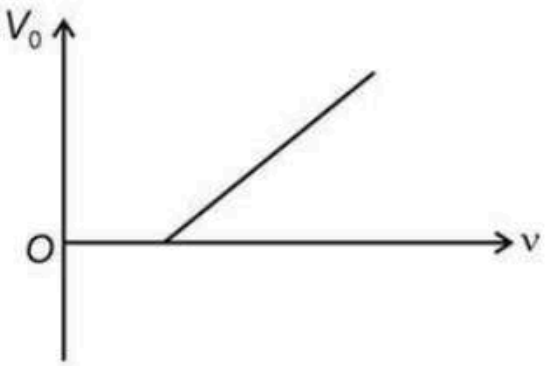
41. A proton and an alpha particle are accelerated from rest by same potential difference. Ratio of their de-Broglie wavelength is

(1) 1 : 1 (2) 2 : 1
(3) 1 : $\sqrt{2}$ (4) $2\sqrt{2}$: 1

42. In Young's double slit experiment, the distance between two slits is halved and the distance between the screen and slit is made three times. Then width of the fringe becomes

(1) Double (2) Four times
(3) Six times (4) Half

43. The stopping potential V_0, for an experiment of photoelectric effect from a metal surface, is plotted with frequency ν of incident light as shown in the figure…



Planck's constant is given by

(1) Slope of the line divided by the charge on an electron
(2) Slope of the line multiplied by the charge on an electron
(3) Intercept cut by the line on *y*-axis
(4) Product of intercept on *y*-axis and charge of the electron

44. An electron in an atom makes transition from the energy state, $E_1 = -\frac{3}{2}E$ to $E_2 = -3E$, then the wavelength of the emitted photon is

(1) $\frac{hc}{3E}$ (2) $\frac{2hc}{3E}$
(3) $\frac{3hc}{2E}$ (4) $\frac{3hc}{E}$

45. In a vernier callipers, (*N* + 1) divisions of vernier scale coincide with *N* divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is

(1) $\frac{1}{10N}$ (2) $\frac{1}{100(N+1)}$
(3) 100*N* (4) 10(*N* + 1)

CHEMISTRY

46. Given below are two statements

Statement I: Due to lanthanoid contraction, Zr and Hf have almost identical radii.

Statement II: Both Ce^{4+} and Yb^{2+} are diamagnetic in nature.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Both statement I and statement II are incorrect
(2) Both statement I and statement II are correct
(3) Statement I is incorrect but statement II is correct
(4) Statement I is correct but statement II is incorrect

47. A solution contains a mixture of Ag^+ (0.1 M) and Hg_2^{2+} (0.1 M) which are to be separated by selective precipitation. The maximum concentration of iodide ion at which only one of them gets precipitated is [Given K_{sp} (AgI) = 8.5×10^{-17}, K_{sp} (Hg_2I_2) = 2.5×10^{-26})]

(1) 8.5×10^{-6} M (2) 5×10^{-13} M
(3) 1.7×10^{-4} M (4) 6×10^{-5} M

48. Consider the following statements

(a) $Cu_2[Fe(CN)_6]$ is chocolate brown in colour
(b) Nickel sulphide dissolves in aqua regia.
(c) The number of five membered ring present in the complex formed by the reaction of Nickel chloride with dimethyl glyoxime in basic medium is four.

Choose the correct statements

(1) (a) and (b) only (2) (b) and (c) only
(3) (a), (b) and (c) (4) (a) and (c) only

49. Which of the following statement is not correct about diborane?

(1) The four terminal hydrogen atoms and the two Boron atoms lie in one plane.
(2) Both the Boron atoms are sp^2 hybridised.
(3) There are two 3-centre-2-electron bonds.
(4) The four terminal B-H bonds are 2-centre-2-electron bonds.

50. Choose the incorrect statement among the following:

(1) $Pt, H_2|HCl(C_1)||HCl(C_2)|Pt, H_2$ represents a cell with $E^{\circ}_{cell} = 0$
(2) On increasing the dilution, specific conductance decreases
(3) On increasing the dilution, molar conductivity decreases
(4) $O_2(g)$ is produced at anode when an aqueous solution of sodium sulphate is electrolysed using inert electrodes

51. Match list I (vitamins) with list II (deficiency diseases)

List I (Vitamins)	List II (Deficiency Diseases)
a. Thiamine	(i) Convulsions
b. Riboflavin	(ii) Beri-beri
c. Pyridoxine	(iii) Scurvy
d. Ascorbic acid	(iv) Cheilosis

Select the correct option.

(1) a(iii), b(i), c(ii), d(iv) (2) a(iii), b(iv), c(i), d(ii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(ii), b(iv), c(i), d(iii)

52. If the hydrogen electrode is dipped in an aqueous solution of pH = 2 at 25°C and 1 bar pressure. The reduction potential of the electrode would be

(1) -0.0591 V (2) +0.0591 V
(3) +0.118 V (4) -0.118 V

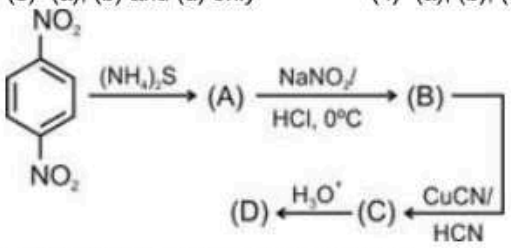
53. Consider the following statements:

(a) Ribose is a monosaccharide pentose sugar
(b) The two six-membered cyclic hemiacetal forms of D–(+)–glucose are called anomers.
(c) Oxidation of glucose with bromine water gives glutamic acid.
(d) Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose.

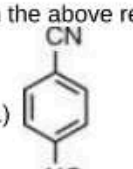
Choose the correct statements.

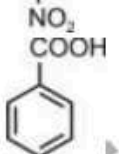
(1) (a), (b) and (c) only (2) (b), (c) and (d) only
(3) (a), (b) and (d) only (4) (a), (b), (c) and (d)

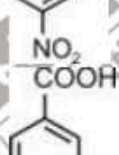
54.

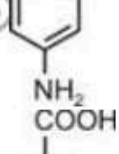


In the above reaction sequence, (D) is

(1)


(2)


(3)


(4)




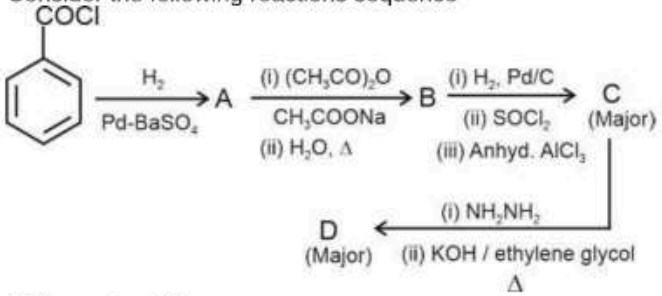
55. Consider the given statements:

(a) Schiff's reagent is rosaniline hydrochloride dye, decolourised by passing SO_2.
(b) Benedict's solution is alkaline $CuSO_4$ complexed with sodium potassium tartarate.
(c) Acetic acid on reaction with MnO at $573\,K$ gives acetone as one of the products.
(d) Formaldehyde on reaction with ammonia gives urotropin.

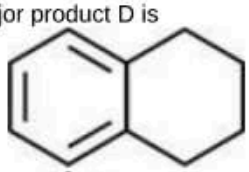
The correct statements are

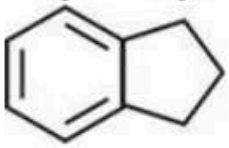
(1) (a) and (d) only (2) (a), (c) and (d) only
(3) (a), (b) and (c) only (4) (a), (b), (c) and (d)

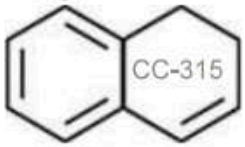
56. Consider the following reactions sequence

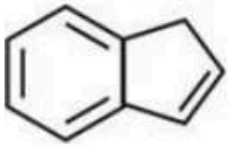


Major product D is

(1) 

(2) 

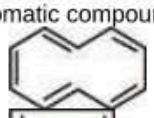
(3) 

(4) 

57. In Dumas method for estimation of nitrogen, 0.1 g of an organic compound gave 30 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is
(Aqueous tension at 300 K = 15 mm Hg)

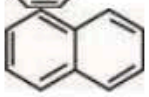
(1) 19.7% (2) 49.1%
(3) 28.2% (4) 31.4%

58. Aromatic compound among the following is

(1) 

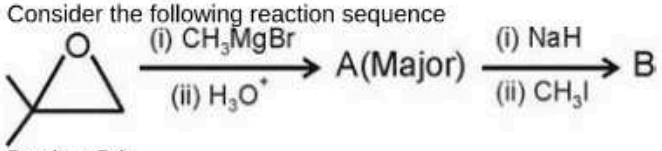
(2) 

(3) 

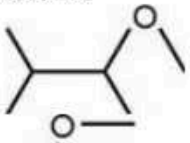
(4) 

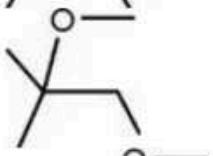
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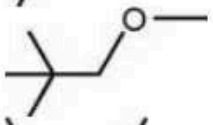
59. Consider the following reaction sequence

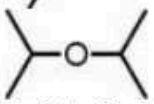


Product B is

(1) 

(2) 

(3) 

(4) 

60. Match list-I with list-II

	List-I		List-II
a.	$[Cr(NH_3)_4Br_2]Br$	(i)	Diamagnetic and exhibits cis-trans isomerism
b.	$[Ti(H_2O)_5Br](NO_3)_2$	(ii)	Diamagnetic and exhibits ionization isomerism
c.	$[Pt(en)(NH_3)Br]NO_3$	(iii)	Paramagnetic and exhibits cis-trans isomerism
d.	$[Co(NH_3)_4(NO_3)_2]NO_3$	(iv)	Paramagnetic and exhibits ionization isomerism

The correct match is

(1) a(iv), b(iii), c(i), d(ii) (2) a(iii), b(iv), c(i), d(ii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iii), b(iv), c(ii), d(i)

61. Number of mole(s) of $KMnO_4$ required to oxidize 1 mole of KI in acidic medium is

(1) 5 (2) 2
(3) $\frac{2}{5}$ (4) $\frac{1}{5}$

62. Consider the following statements:

(a) Griss-Ilosvay test is the confirmatory test of the nitrite ion
(b) Brown ring test is the confirmatory test of nitrate ion
(c) In qualitative analysis, group reagent for Ba^{2+} and Ni^{2+} cations are $(NH_4)_2CO_3$ in presence of NH_4OH and H_2S gas in presence of dilute HCl respectively
(d) On passing ammonia gas through Nessler's reagent a brown precipitate is formed
(e) Lead sulphate is soluble in ammonium acetate solution due to the formation of ammonium tetraacetoplumbate(II)

The correct statements are

(1) (b), (d) and (e) only (2) (a), (b), (d) and (e) only
(3) (a) and (c) only (4) (a), (b), (c), (d) and (e)

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63. Given below are two statements one is labelled as assertion (A) other is labelled as reason (R).

Assertion (A): O_2 is a paramagnetic species.

Reason (R): O_2 has two unpaired electrons in π-antibonding molecular orbital.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

64. Consider the following statements:

Statement I: Energy of 2s orbital of hydrogen atom is less than that of 2s orbital of lithium atom.

Statement II: The number of angular nodes in 3p orbital is one.

In the light of above statements, choose the correct option.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

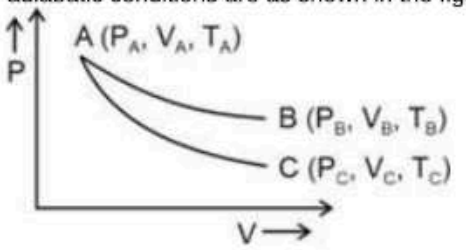
(4) Statement I is incorrect but statement II is correct

65. Choose the correct oxidation state(s) of sulphur atoms in tetrathionate ion

(1) +6 and 0 only (2) +6 only

(3) +5 and 0 only (4) +5 and +6 only

66. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



a. $\Delta U_{isothermal} < \Delta U_{adiabatic}$ (Magnitude only)

b. $T_C > T_B$

c. $T_B = T_A$

d. $w_{isothermal} < w_{adiabatic}$ (By system)

Identify the incorrect statements.

(1) a, b and c only (2) b and c only

(3) b and d only (4) a, b and d only

67. Given below are two statements:

Statement I: XeF_2 on reaction with PF_5 gives $[XeF]^+[PF_6]^-$.

Statement II: XeF_6 does not react with NaF.

In the light of the above statements, choose the most appropriate answer from the options given below.

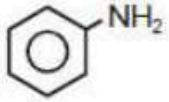
(1) Statement I is correct but statement II is incorrect

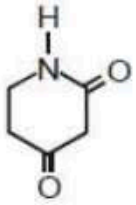
(2) Statement I is incorrect but statement II is correct

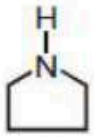
(3) Both statement I and statement II are correct

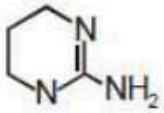
(4) Both statement I and statement II are incorrect

68. Which of the following is the strongest base?

(1) 

(2) 

(3) 

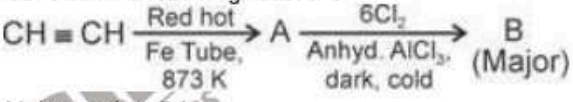
(4) 

69. The element with the least negative value of electron gain enthalpy among the following elements is

(1) F (2) S

(3) Se (4) O

70. Consider the following reaction.



Major product B is

(1) $C_6H_6Cl_6$ (2) $C_6H_3Cl_3$

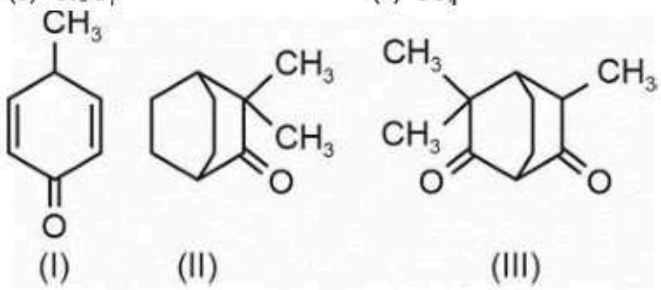
(3) $C_6H_4Cl_2$ (4) C_6Cl_6

71. Among the following species, the one that can undergo disproportionation reaction is

(1) ClO_2^- (2) MnO_4^-

(3) $Cr_2O_7^{2-}$ (4) SO_4^{2-}

72.



Which of the given compounds can exhibit tautomerism?

(1) I, II and III (2) Only I and II

(3) Only I and III (4) Only II and III

73. Consider the following statements:

(a) A catalyst does not alter Gibbs energy, ΔG of a reaction.

(b) Thermal decomposition of HI on gold surface is an example of zero order reaction.

(c) All natural and artificial radioactive decay of unstable nuclei take place by first order kinetics.

(d) A catalyst catalyses both spontaneous and non-spontaneous reactions.

The correct statements are

(1) (a), (b) and (c) only (2) (b), (c) and (d) only

(3) (a) and (b) only (4) (a), (b), (c) and (d)

74. The entropy change when 1 mole of ideal gas at 27°C is expanded reversibly from 2 litre to 20 litre is

(1) 2 cal K^{-1} mol^{-1} (2) 4.6 cal K^{-1} mol^{-1}

(3) 4 J K^{-1} mol^{-1} (4) 4.6 kJ K^{-1} mol^{-1}

75. In which of the following concentration cells, cell reaction will occur spontaneously?

(1) Zn(s) | Zn^{2+}(0.1 M) || Zn^{2+}(0.01 M) | Zn(s)

(2) Zn(s) | Zn^{2+}(0.2 M) || Zn^{2+}(0.02 M) | Zn(s)

(3) Zn(s) | Zn^{2+}(0.2 M) || Zn^{2+}(0.2 M) | Zn(s)

(4) Zn(s) | Zn^{2+}(0.01 M) || Zn^{2+}(0.1 M) | Zn(s)

76. Consider the given reaction

A + 2B → 3C

2 moles of each A and B have been taken in a closed container to form C.

Consider the following observations.

(a) A is limiting reagent

(b) 3 moles of C is formed

(c) 1 mole of B is left after the reaction.

The incorrect statement(s) is/are

(1) (b) only (2) (a) and (c) only

(3) (a) and (b) only (4) (b) and (c) only

77. The correct order of first ionization enthalpy for the given elements is

(1) Be < B < C < N < O < F (2) B < Be < C < O < N < F

(3) Be < B < C < O < N < F (4) B < Be < C < N < O < F

78. The primary and secondary valencies of Co metal ion in $[Co(NH_3)_4Cl_2]Cl$ respectively are

(1) 3 and 6 (2) 3 and 3

(3) 6 and 6 (4) 6 and 3

79. Which experimental phenomenon is explained by wave nature of electromagnetic radiation?

(1) Diffraction

(2) Black-body radiation

(3) Line spectra of hydrogen atom

(4) Photoelectric effect

80. Given below are two statements: one is labelled as **Assertion (A)** the other is labelled as **Reason (R).**

Assertion (A): Upon addition of $HgCl_2$(aq) to the following reaction mixture:

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$;

The intensity of red colour decreases.

Reason (R) : Hg^{2+} reacts with SCN^- to form stable complex ion $[Hg(SCN)_4]^{2-}$.

In the light of above statements, select the most appropriate option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

81. The given table has values of Henry's law constant for some gases in water at 293 K.

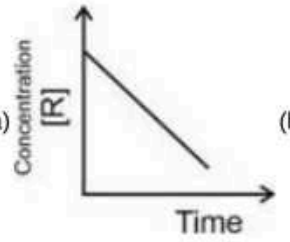
Gas	K_H (Kbar)
He	144.97
H_2	69.16
N_2	76.48
O_2	34.86

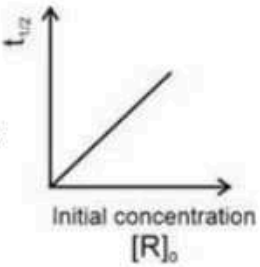
The correct order of solubility of given gases in water is

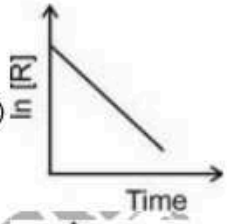
(1) He < H_2 < N_2 < O_2 (2) He < N_2 < H_2 < O_2

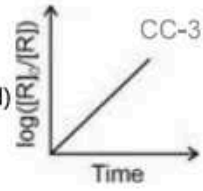
(3) O_2 < N_2 < H_2 < He (4) N_2 < O_2 < H_2 < He

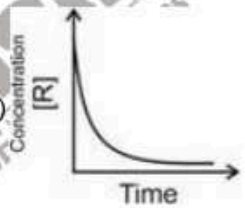
82. For the following graph:

(a) 

(b) 

(c) 

(d) 

(e) 

Choose from the options given below, the correct one regarding order of reaction

(1) (a) is zero order; (b) and (e) are first order

(2) (a) and (d) are first order; (b) is zero order

(3) (d) and (e) are zero order; (a) and (b) are first order

(4) (a) and (b) are zero order; (c) and (e) are first order

83. Which among the following complexes will show Jahn Teller distortion?

(a) $[FeF_6]^{3-}$

(b) $[CoF_6]^{3-}$

(c) $[Co(C_2O_4)_3]^{3-}$

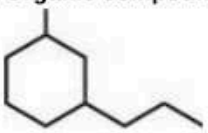
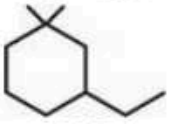
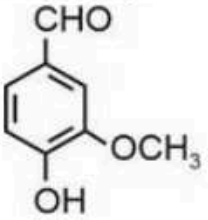
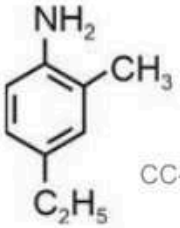
(d) $[Fe(CN)_6]^{3-}$

Choose the correct option.

(1) (a) and (c) only (2) (a), (b) and (c) only

(3) (b) and (d) only (4) (a), (b), (c) and (d)

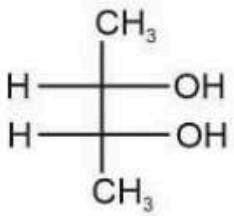
84. Choose the incorrect match.

	Organic compound	IUPAC name
(1)		1-Methyl-3-propylcyclohexane
(2)		1-Ethyl-3,3-dimethylcyclohexane
(3)		4-Hydroxy-3-methoxy benzaldehyde
(4)		4-Ethyl-2-methylaniline

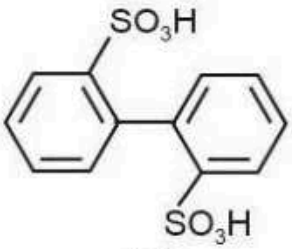
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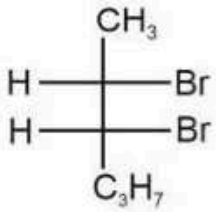
(1) (1) (2) (2)
(3) (3) (4) (4)

85. Consider the following compounds:


(a)


(b)

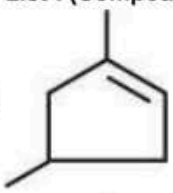
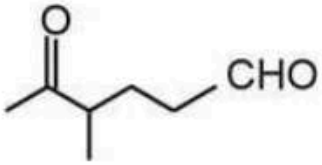
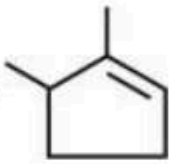
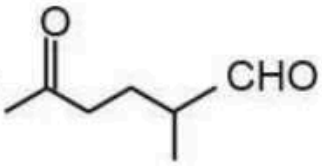
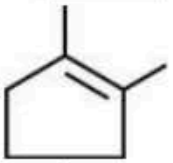
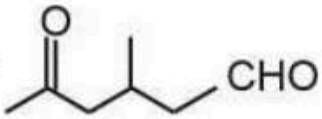
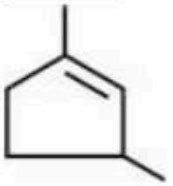

(c)


(d)

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Optically active compound(s) is/are
(1) (a), (b), (c) and (d) (2) (b), (c) and (d) only
(3) (b) and (c) only (4) (b) only

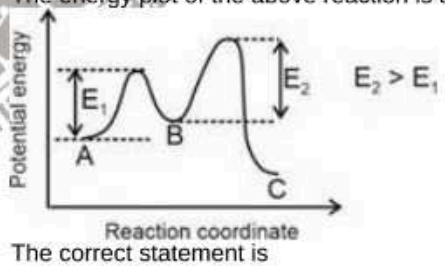
86. Match List-I with List-II

List-I (Compound)	List-II (Ozonolysis products)
(a) 	(i) 
(b) 	(ii) 
(c) 	(iii) 
(d) 	(iv) 

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The correct match is
(1) a(i), b(iii), c(ii), d(iv) (2) a(iii), b(i), c(iv), d(ii)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iii), c(i), d(iv)

87. A → B → C
The energy plot of the above reaction is as follows:



$E_2 > E_1$

The correct statement is
(1) A → B is rate determining step
(2) The reaction is endothermic
(3) It is an example of elementary reaction
(4) B → C is the slowest step of reaction

88. The chemical reaction, $2O_3 \longrightarrow 3O_2$ proceeds as follows:
(i) $O_3 \rightleftharpoons O_2 + O$(fast)
(ii) $O + O_3 \rightleftharpoons 2O_2$(slow)
The unit of rate constant of reaction $2O_3 \longrightarrow 3O_2$, is
(1) mol $L^{-1}s^{-1}$ (2) s^{-1}
(3) L $mol^{-1}s^{-1}$ (4) L^2 mol^{-2} s^{-1}

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89. Match the List I with List II

List I (Compounds)		List II (Method of preparation)
a. Metaphosphoric acid	(i)	$P_2O_3 + H_2O$
b. Phosphonic acid	(ii)	$PCl_3 + H_3PO_3$
c. Hypophosphoric acid	(iii)	Red P_4 + alkali
d. Pyrophosphorous acid	(iv)	Phosphorus acid + Br_2, heat in a sealed tube

The correct match is
(1) a(ii), b(iii), c(i), d(iv)
(2) a(iv), b(i), c(iii), d(ii)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(iv), b(i), c(ii), d(iii)

90. Two solids dissociate in a container as follows:

A(s) $\rightleftharpoons$ B(g) + C(g); K_{p_1} = a atm^2

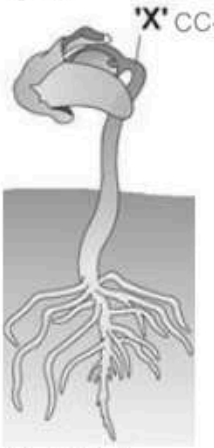
D(s) $\rightleftharpoons$ C(g) + E(g); K_{p_2} = b atm^2

The total equilibrium pressure when both the solids dissociate simultaneously is:
(1) (a + b) atm
(2) $2\left(\sqrt{a+b}\right)$ atm
(3) (a + b)^2 atm
(4) $\frac{a+b}{\sqrt{2}}$ atm

BIOLOGY

91. Pyrenoids are
(1) Storage bodies in most of the green algae
(2) Double membrane bound cell organelles
(3) Protein synthesising bodies in all the members of algae
(4) Complex carbohydrates in the cell wall of algae

92. Select the **correct** option for the label 'X' in the following figure.



(1) Cotyledon
(2) Seed coat
(3) Hypocotyl
(4) Epicotyl hook

93. The basal part of the ovule is represented by
(1) Funicle
(2) Hilum
(3) Micropyle
(4) Chalaza

94. Match column I with column II and select the **correct** option.

Column-I (Biological name)		Column-II (Taxonomic categories)
a. *Musca domestica*	(i)	Poales
b. *Mangifera indica*	(ii)	Polymoniales
c. *Solanum melongena*	(iii)	Diptera
d. *Triticum aestivum*	(iv)	Sapindales

(1) a(iii); b(iv); c(ii); d(i)
(2) a(iii); b(iv); c(i); d(ii)
(3) a(iv); b(iii); c(ii); d(i)
(4) a(iv); b(iii); c(i); d(ii)

95. Identify the **incorrect** match from the following
(1) Auxin – NAA
(2) Gibberellins – Indole compound
(3) Ethylene – Gaseous plant hormone
(4) Cytokinin – Present in coconut milk

96. In both lactic acid and alcohol fermentation:
(1) CO_2 and ethanol are produced
(2) Reducing agent NADH + H^+ is re-oxidised to NAD^+
(3) Large amount of ATP is produced
(4) Pyruvate decarboxylase is used

97. Read the following statements and select the **correct** option.
Statement A : All living organisms retain the enzymatic machinery for glycolysis.
Statement B : In naturally fermented alcoholic beverages, maximum concentration of alcohol could be less than 13%.
(1) Only A is incorrect
(2) Only B is incorrect
(3) Both A and B are correct
(4) Both A and B are incorrect

98. Who was the author of the book, Anatomy of seed plants?
(1) Ernst Mayr
(2) Katherine Esau
(3) Peter Raven
(4) Ramachandran

99. Select the **incorrectly** matched pair
(1) Mutualism – Fig and fig wasp
(2) Parasitism – Copepods on marine fishes
(3) Commensalism – Epiphytes growing on mango
(4) Amensalism – *Cuscuta* on hedge plants

100. Species-Area Relationship was first proposed by
(1) Edward Wilson
(2) Alexander von Humboldt
(3) Paul Ehrlich
(4) Robert May

101. Perisperm is found in the seeds of
(1) Castor and bean
(2) Pea and bean
(3) Black pepper and beet
(4) Maize and castor

102. How many of the following alcoholic beverages are produced without distillation?
Whisky, Brandy, Rum, Wine, Gin, Vodka, Beer
(1) 3
(2) 5
(3) 2
(4) 4

103. An *E. coli* cell containing hybrid DNA for isotopes of nitrogen element is allowed to multiply in a medium containing $^{14}NH_4Cl$. What percent of DNA will be hybrid in 2nd generation?
(1) 50%
(2) 75%
(3) 25%
(4) 12.5%

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104. Identify the **correct** floral formula of a family to which belladonna belongs.

(1) $Br \oplus \female P_{(3+3)} A_{3+3} \underline{G}_{(3)}$

(2) $\oplus \female K_{(5)} C_{(5)} A_{5} \underline{G}_{(2)}$

(3) $\% \female K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G}_{1}$

(4) $\oplus \female K_{2+2} C_{x4} A_{2+4} \underline{G}_{(2)}$

105. Match column I with column II.

	Column I		Column II
a.	Bulliform cells	(i)	Composed of dumb-bell shaped guard cells and tiny pore in the leaves of grasses
b.	Cuticle	(ii)	Usually single layered and present in most plant parts
c.	Stomata	(iii)	Large, empty and colourless cells found in monocots
d.	Epidermis	(iv)	Waxy thick layer prevents water loss

Choose the **correct** option.

(1) a(ii), b(iii), c(iv), d(i) (2) a(iii), b(iv), c(i), d(ii)
(3) a(i), b(ii), c(iii), d(iv) (4) a(iv), b(i), c(ii), d(iii)

106. Consider the following w.r.t. epithelial tissue.

(a) The free surface of epithelial tissue always faces the outside environment.

(b) The cells are compactly packed with large intercellular matrix.

(c) The ciliated epithelia are mainly present in the inner surface of hollow organs like bronchioles.

(d) The ciliated epithelium moves the particles or mucus in a specific direction over the epithelium.

Select the option that represents **incorrect** statement(s) only

(1) (a), (b), (c) and (d) (2) (a), (b) and (c) only
(3) (a) and (b) only (4) (a) only

107. Choose the **correct** statement.

(1) All the different types of epithelial tissues have their own blood supply.
(2) All the cells associated with the neural system are excitable cells.
(3) All unicellular organisms exhibit division of labour.
(4) All four different types of animal tissues are present in our heart.

108. Which of the following is **not** a polymer?

(1) 'X' fibres that act as intercellular ground substance and provide strength to the connective tissue
(2) 'P' that are esters of fatty acids with alcohol
(3) 'Y' that is involved in storage and expression of genomic information
(4) 'Z' that catalyses the hydrolysis of proteins

109. Comprehend the given statements.

Statement A: There is one feature common to all those compounds found in the acid-soluble pool, *i.e.*, they have molecular weights ranging from 18 to around 800 daltons (Da) approximately.

Statement B: Chemical compounds found in living organisms which have molecular weight ranging from 10,000 Da and above are found in acid-insoluble fraction.

Statement C: Molecular weights of lipids do not exceed 800 Da but they are found in the acid-insoluble fraction.

Select the **correct** option.

(1) Statements A and C are correct but B is incorrect
(2) Statements B and C are correct but A is incorrect
(3) Statements A, B and C are correct
(4) Statements A and B are correct but C is incorrect

110. The most abundant protein present in the animal world is 'P'.

Select the **correct** option for 'P'.

(1) Also found in the blood
(2) Arranged in an irregular pattern in dense connective tissue that is present beneath the skin
(3) Not stabilized by hydrogen bonds
(4) A storage homopolymer in plants

111. Read the given statements and select the correct option.

Statement A: The type of co-factor found in peroxidase is actually an organic compound which is tightly bound to the apoenzyme.

Statement B: Catalase and carboxypeptidase possess same type of co-factor.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

112. Which of the following statements is/are **incorrect**?

a. The presence of competitive inhibitor decreases the K_m of the enzyme for the substrate.

b. In non-competitive inhibition, the enzyme molecule is structurally changed by the inhibitor.

c. Competitive inhibition prevents the formation of the enzyme-substrate complex, indicating a reduction in binding affinity.

d. The inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition.

Choose the **correct** option.

(1) a and d (2) b and c
(3) d only (4) a only

113. Given below are the features of a respiratory regulatory centre called 'X'.

(a) Neural signals from this centre can reduce the duration of inspiration.

(b) It can moderate the functions of respiratory rhythm centre.

(c) It is present in hind brain.

Select the option that correctly identifies 'X'.

(1) Pneumotaxic centre (2) Hunger centre
(3) Chemosensitive area (4) Association area

114. Vital capacity can be represented as

(a) TLC – RV
(b) IRV + TV + ERV
(c) ERV + RV

Choose the **correct** option.

(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

115. Select the **incorrect** option w.r.t. ECG.
(1) To obtain a standard ECG, a patient is connected to the machine with three electrical leads, one to each wrist and to the chest region
(2) The ST segment represents the interval between the ventricular depolarization and repolarization
(3) Increase electrical activity of ventricles may lead to an increase in the height of QRS complex.
(4) The 'P' wave indicates positive deflection from the baseline and it represents atrial depolarization

116. In a dihybrid cross of true-breeding round and yellow seeded pea plant with wrinkled green seeded pea plants, what will be the percentage of recombinants in F_2 generation?
(1) 60% (2) 25%
(3) 37.5% (4) 50%

117. Read the following statements and choose the **correct** ones.
A. Carotenoids protect chlorophyll *a* from photo-oxidation.
B. Dark reactions are not directly light driven.
C. Chlorophyll *a* molecule shows maximum absorption in red and blue regions of the visible spectrum of light.
D. Light harvesting complexes are composed of hundreds of pigment molecules bound to lipids only.
(1) A, B & C (2) B, D & C
(3) A, C & D (4) D, B & A

118. Which of the following is a **correct** statement w.r.t. alien species invasion?
(1) *Clarias gariepinus* was introduced in India accidently
(2) *Lantana* has become a serious weed in multiple wetlands in India
(3) Cichlid fish was introduced in Lake Victoria and has killed multiple species of Nile perch
(4) *Parthenium* is a terrestrial weed which came to India with imported wheat

119. Which is **incorrect** statement w.r.t. biocontrol agents?
(1) Baculoviruses are pathogens that attack insects and other arthropods
(2) Ladybird and dragonflies are useful to get rid of nematodes
(3) *Bacillus thuringiensis* was introduced to control butterfly catterpillars
(4) *Trichoderma* is a fungus used against a variety of plant pathogens

120. Drones do not have (i) _____ and thus cannot have (ii) _____, but have a (iii) _____ and can have (iv) _____.
The correct words for (i), (ii), (iii) and (iv) respectively are
(1) Grandfather, grandsons, father and sons
(2) Grandsons, sons, father and grandfather
(3) Sons, grandsons, father and grandfather
(4) Father, sons, grandfather and grandsons

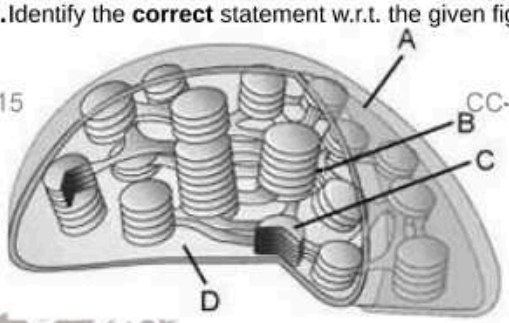
121. Match column I with column II and choose the **correct** option.

	Column I		Column II
a.	Metacentric	(i)	Centromere of chromosome is situated close to one of its end.
b.	Acrocentric	(ii)	They appear "L-shaped" during anaphase
c.	Sub metacentric	(iii)	Chromosome appears to have a single arm
d.	Telocentric	(iv)	It is also called isobrachial chromosome

(1) a(ii), b(iv), c(i), d(iii) (2) a(iv), b(i), c(ii), d(iii)
(3) a(iii), b(i), c(iv), d(ii) (4) a(iv), b(iii), c(i), d(ii)

122. If a cell in the G_1 phase has 10 chromosomes and 5 picograms of DNA, what would be the corresponding chromosome number and DNA content in the G_2 phase?
(1) 10 chromosomes; 10 pg (2) 20 chromosomes; 5 pg
(3) 5 chromosomes; 20 pg (4) 5 chromosomes; 10 pg

123. Identify the **correct** statement w.r.t. the given figure.



A. It is more permeable in comparison to the inner membrane.
B. These are stacked flat membranous tubules arranged like piles of coins which help in CO_2 fixation.
C. These are organised flattened membranous sacs present in the stroma.
D. It has enzymes for synthesis of sugar.
(1) A and B only (2) B and C only
(3) A, C and D (4) B, C and D

124. Match the **column I** with **column II** and select the **correct** option.

Column I		Column II
a. RER	(i)	Site of formation of glycoproteins
b. SER	(ii)	Rich in hydrolytic enzymes
c. Lysosomes	(iii)	Provide precursors of enzymes for the formation of lysosomes
d. Golgi bodies	(iv)	Detoxification of drugs

(1) a(iii), b(iv), c(ii), d(i) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(ii), b(iii), c(i), d(iv)

125. Select the **incorrect** statement regarding Golgi apparatus.
(1) This cell organelle was named by Camillo Golgi
(2) Its *cis* face is the forming face and *trans* face is the maturing face
(3) Its *cis* and *trans* faces are interconnected
(4) It consists of many cisternae stacked parallel to each other

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126. Read the following statements and state them as True (**T**) or False (**F**) and select the **correct** option.

A. By the end of prophase of mitosis, diakinesis and prophase II, the nucleolus disappears and the nuclear envelope breaks down.

B. Cytokinesis follows telophase I and results in the formation of tetrad of cells.

C. Equational division is the mechanism by which conservation of the specific chromosome number of each species is achieved across generations in sexually reproducing organisms.

D. Interkinesis is followed by the prophase I.

(1) A-(T), B-(F), C-(T), D-(F)
(2) A-(T), B-(T), C-(T), D-(T)
(3) A-(F), B-(T), C-(F), D-(T)
(4) A-(T), B-(F), C-(F), D-(F)

127. Cytokinesis of plant cells differs from animal cells as the former

(1) Occurs by formation of furrow
(2) Starts at periphery
(3) Occurs by formation of cell plate
(4) Occurs before karyokinesis

128. Racemose inflorescence differ from cymose inflorescence as the latter lacks

(1) Unlimited growth of inflorescence axis
(2) Arrangement of flowers in basipetal order
(3) Older flowers at top
(4) Flower on tip of main axis

129. How many of the given features are related to maize seed?

a. Scutellum
b. Albuminous
c. Thick seed coat
d. Radicle enclosed in coleoptile
e. Pericarp fuses with seed coat

(1) 3
(2) 4
(3) 5
(4) 2

130. All are required during chemiosmosis process for ATP synthesis, **except**

(1) Membrane
(2) Proton gradient
(3) NADP reductase
(4) Proton pump

131. Neutrophils and monocytes differ from each other in all of the following **except**

(1) The shape of their nuclei in mature form
(2) Their contribution to total WBC content of blood
(3) Their mode of action against foreign organisms
(4) Their cytoplasmic contents

132. Which of the following statement is incorrect w.r.t. juxtamedullary nephrons?

(1) Juxtamedullary nephrons have the renal corpuscles in the inner cortex near the corticomedullary junction
(2) These nephrons have long loop of Henle
(3) The tubular portion of juxtamedullary nephrons is supplied by the straight capillaries called vasa recta
(4) They make about 80% of the total nephrons

133. The proximity between the Henle's loop and vasa recta, as well as counter current in them help in maintaining an increasing osmolarity towards the inner medullary interstitium. From 300 mosmL^{-1} in cortex, the osmolarity of the glomerular filtrate in the hair pin loop of Henle is about

(1) Two times concentrated than the filtrate found in PCT
(2) Four times concentrated than the filtrate found in DCT
(3) Four times concentrated than the filtrate found in PCT
(4) Two times concentrated than the filtrate found in DCT

134. Choose the feature which is **not** associated with white muscle fibres

(1) Lighter in colour
(2) Fast rate of contraction but for a short period
(3) Dependence mainly on glycolysis for energy production and soon get fatigued
(4) Lesser sarcoplasmic reticulum as compared to red muscle fibres

135. The **correct** sequence of events at the neuromuscular junction is

a. Uptake of Ca^{2+} into the presynaptic terminals
b. Action potential in the motor nerve
c. Release of acetyl-choline (Ach)
d. Depolarisation in the muscle end plate
e. Interaction between Ach and Ach-specific receptors

(1) a, b, c, d & e
(2) a, b, d, c & e
(3) b, a, c, d & e
(4) b, a, c, e & d

136. Match column-I with column-II and choose the **correct** option

	column-I		**column-II**
a.	Thin fibrous membrane, holding thick filaments together	(i)	Z-Disc
b.	Functional unit of contraction	(ii)	A-band
c.	Synapse between a motor neuron & a muscle fibre	(iii)	M-line
d.	Area in the sarcomere where both thick & thin filaments are found	(iv)	Sarcomere
		(v)	Neuromuscular junction

(1) a(i), b(iv), c(v), d(iii)
(2) a(iii), b(ii), c(v), d(i)
(3) a(iii), b(iv), c(v), d(ii)
(4) a(i), b(ii), c(iii), d(iv)

137. In humans, the skull is dicondylic and can perform the "nodding" movements due to the presence of

(1) Two occipital condyles that articulate with the facets of the atlas vertebra
(2) Three occipital condyles that articulate directly with the axis vertebra
(3) Gliding joint between the atlas and odontoid process of axis
(4) Ball and socket joint between atlas and axis vertebrae

138. Select the **incorrect** option w.r.t. neurons.

(1) They are linked with other cells of the body to form complex neural connections for transmission of information
(2) The plasma membrane of the short fibres which arise from cell body contain numerous receptor sites for binding of chemical messengers
(3) The proximal end of the long fibre attached to cell body is branched, where each branch terminates as fusiform-shaped structure
(4) Terminal branch of an axon is unmyelinated

139. Many subconscious activities of our body are controlled by the

(a) Medulla oblongata
(b) Pons
(c) Hypothalamus
(d) Cerebellum
(e) Amygdala

Select the **correct** option.

(1) (c), (d) and (e) only
(2) (a), (b), (c) and (d) only
(3) (a), (b), (c), (d) and (e)
(4) (b), (d) and (e) only

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140. Consider the following w.r.t. glucagon
(a) It is secreted from the alpha cells of pancreas.
(b) Stimulates the process of gluconeogenesis.
(c) It acts mainly on hepatocytes and stimulates glycogenolysis.
(d) High blood glucose level promotes its release *via* negative feedback mechanism.
How many of the above is/are **correct**?
(1) One (2) Three
(3) Two (4) Four

141. The hormone 'X' is released by the four small glands located on the backside of the 'Y' gland.
Identify 'X' and 'Y' w.r.t. humans and choose the **correct** option.
(1) Excess secretion of 'X' can lead to tetany
(2) 'Y' has follicles that are lined by columnar epithelium
(3) Kidneys lack target cells for hormone 'X' and hormone of 'Y'
(4) 'X' stimulates osteoclast activity and bone resorption

142. Metameric segmentation is seen in members of all of the given phyla, **except**
(1) Annelida (2) Arthropoda
(3) Hemichordata (4) Chordata

143. Match column I with column II w.r.t. excretory structures in various animals.

	Column I		**Column II**
a.	Earthworm	(i)	Green glands
b.	*Balanoglossus*	(ii)	Kidneys
c.	Vulture	(iii)	Proboscis gland
d.	*Palaeomon*	(iv)	Nephridia

(1) a(i), b(ii), c(iii), d(iv) (2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iv), b(iii), c(ii), d(i)

144. Consider the statements A and B and select the **correct** option.
Statement A: All gnathostomes are not vertebrates but all vertebrates are gnathostomes.
Statement B: All tetrapods are gnathostomes, but all gnathostomes are not tetrapods.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

145. Which of the following organisms is **correctly** described?

(1)	*Rana*	–	Moist skin, lives in or near water
(2)	*Psittacula*	–	Warm blooded, 3-chambered heart
(3)	*Balaenoptera*	–	Sexes are separate, fertilisation is external
(4)	*Chelone*	–	Poikilothermic, development is indirect

(1) (1) (2) (2)
(3) (3) (4) (4)

146. Read the following w.r.t. cockroaches:
(a) In both the sexes, 10th abdominal segment bears a pair of anal style.
(b) In female cockroaches, anterior part of brood pouch contains gonopore, spermathecal pore and opening of collaterial glands.
(c) Hypopharynx acts as tongue, lies within the cavity enclosed by the mouthparts.
(d) Maxillae possess grinding and incising regions.
How many of the above statements is/are **correct**?
(1) Three (2) Two
(3) Four (4) Zero

147. Identify the **incorrect** matches w.r.t. frogs.
a. HCl and gastric juice – Secreted from walls of intestine
b. Bile and pancreatic juice – Synthesised and secreted from common bile duct
c. Villi and microvilli – Finger-like folds in stomach
d. Chyme – Partially digested food in stomach
Select the **correct** option.
(1) a, b and c (2) b, c and d
(3) a, c and d (4) a, b and d

148. **Assertion(A):** The foregut of cockroach does not permit significant absorption of nutrients.
Reason(R): The entire foregut is lined by tough, impermeable cuticle and food substances are completely digested upto absorbable forms in intestine.
In the light of above statements, choose the **correct** option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

149. Choose the **correct** match w.r.t. humans.

(1)	Oogenesis	–	Initiated before the birth of females
(2)	Acrosome	–	Filled with hormones
(3)	Antrum	–	A cavity present within primary follicles
(4)	First polar body	–	Larger than secondary oocyte in size

(1) (1) (2) (2)
(3) (3) (4) (4)

150. Select the correct combination of hormones that are produced by temporary endocrine gland found attached with uterus.
(1) hCG, hPL, progesterone, relaxin and luteinising hormone
(2) hCG, hPL, thyrocalcitonin and progesterone
(3) hCG, prolactin, hPL, estrogen and thymosin
(4) hCG, hPL, progesterone and estrogen

151. Select the **incorrect** statement w.r.t. humans.
(1) For normal fertility, out of 200 million ejaculated sperms, atleast 120 million sperms must have normal shape and size.
(2) The release of spermatozoa from Sertoli cells into the lumen of the seminiferous tubules is called spermiation.
(3) The spermatogenesis is a continuous process that occurs throughout life, under the influence of testosterone, which is released at a constant rate in a pulsatile manner throughout life.
(4) Pulse frequency of testosterone in blood regulates the secretion of GnRH from the hypothalamus which thereby regulates secretion of FSH and LH from the anterior pituitary.

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152. Select the incorrect statement w.r.t. post fertilization events.
(1) After the formation of zygote, cleavage occurs, which is characterised by division of cells without change in their ploidy.
(2) Placenta forms the functional connection between mother and the foetus
(3) Level of progesterone remains high in order to maintain the uterine wall till parturition
(4) There is completion of second meiotic division of the secondary oocyte and formation of second polar body and a haploid ovum

153. Choose the incorrect option w.r.t ICSI.
(1) Sperm is placed inside the egg with a microscopic needle
(2) The embryo formed by this procedure is transferred to the uterus
(3) The semen collected from a healthy donor is introduced in the vagina of the female
(4) It falls under the umbrella of IVF technology

154. A woman using steroidal oral contraceptive pills shows the following hormonal profile:
- Suppressed LH surge
- Reduced FSH
- Normal estrogen fluctuation absent
- Thick cervical mucus
- Endometrium unsuitable for implantation

Which of the following mechanisms may be hampered to prevent the pregnancy in the above mentioned case?
(a) Fertilisation
(b) Ovulation
(c) Implantation
Choose the **most appropriate** option.
(1) Only (a) (2) Only (a) and (b)
(3) Only (c) (4) (a), (b) and (c)

155. The embryological support for evolution, proposed by Ernst Heckel on careful study, was disapproved by
(1) Alfred Wallace (2) Lamarck
(3) Karl Ernst von Baer (4) Louis Pasteur

156. Select the **odd** one w.r.t. red algae.
(1) *Polysiphonia* (2) *Porphyra*
(3) *Gelidium* (4) *Dictyota*

157. Given below are the two statements: one is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A): *Cycas* and *Marchantia*, both are dioecious belonging to different groups of plants.
Reason (R): In gymnosperms and bryophytes, the male and female sex organs are always borne on different plants.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

158. Select the **incorrectly** matched pair.
(1) *Albugo* – Parasitic fungi on mustard
(2) *Rhizopus* – Aseptate and coenocytic mycelium
(3) *Puccinia* – Asexual reproduction is mostly by conidia
(4) *Agaricus* – Karyogamy and meiosis take place in the basidium

159. Sleeping sickness and bovine spongiform encephalopathy are respectively caused by
(1) *Trypanosoma* and prions
(2) *Plasmodium* and viroids
(3) *Entamoeba* and viruses
(4) *Paramoecium* and virusoids

160. Mark the given statements as **True(T)** or **False(F)** and select the **correct** option.
a. Deletions and insertions of DNA cause substitution mutation.
b. UV rays can act as mutagens.
c. Cancer cells commonly show chromosomal abnormalities

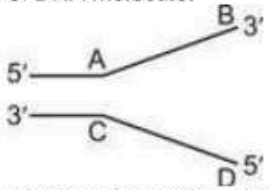
	a	**b**	**c**
(1)	T	T	F
(2)	T	F	T
(3)	F	T	T
(4)	F	F	T

(1) (1) (2) (2)
(3) (3) (4) (4)

161. Among the following crosses regarding ABO blood groups in human being, select the ones that show same number of different genotypes as that of phenotypes in the progeny.
(a) $I^A i \times I^B i$
(b) $I^B I^B \times I^A i$
(c) $I^A I^B \times I^B i$
(d) $I^A I^A \times I^A I^B$
Choose the **correct** option.
(1) (b) and (d) only (2) (a), (b) and (d) only
(3) (a), (c) and (d) only (4) (a) and (c) only

162. Phosphodiester linkage is present between
(1) Two adjacent nucleotides in DNA
(2) Two sugar moieties of two adjacent DNA strands
(3) Nitrogenous base and sugar moiety
(4) Two nitrogenous bases of opposite DNA strands

163. Consider the following replication fork formed by unwinding of DNA molecule.



Choose the option that indicates the direction in which replication is continuous.
(1) A → B (2) B → A
(3) C → D (4) D → C

164. If the amount of thymine is 20% of the total nitrogenous bases in DNA of *E. coli*, then the percentage of cytosine will be
(1) 20% (2) 25%
(3) 30% (4) 40%

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165. Read the following **Assertion (A)** and **Reason (R)** statements.

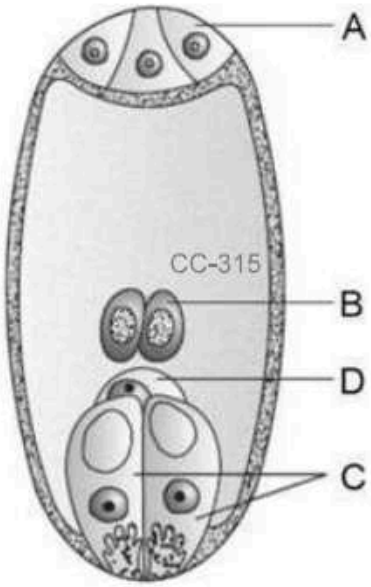
Assertion (A) : In prokaryotes, the translation is not possible if methyl guanosine triphosphate is not found at 5' end of mRNA.

Reason (R) : In prokaryotes, methyl guanosine triphosphate at the 5' end is identified by 16S rRNA of ribosomal unit.

In the light of above statements select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

166. Observe the figure given below.



Choose the **correct** option from the following for 'A', 'B', 'C', 'D'.

	A	B	C	D
(1)	Synergids	Antipodals	Egg	Central cell
(2)	Synergids	Polar nuclei	Antipodals	Egg
(3)	Antipodals	Egg	Polar nuclei	Synergids
(4)	Antipodals	Polar nuclei	Synergids	Egg

(1) (1) (2) (2)
(3) (3) (4) (4)

167. Pyramid of energy is

(1) Always upright (2) Inverted
(3) Urn-like (4) Spindle-shaped

168. Vertical distribution of different species occupying different levels in an ecosystem is called

(1) Species composition (2) Stratification
(3) Standing crop (4) Species diversity

169. The type of population interaction exhibited by Abingdon tortoise and goats is

(1) Competition (2) Commensalism
(3) Amensalism (4) Parasitism

170. Read the given statements (a-d).

a. Methanogens are present in the gut of several ruminant animals.
b. Chemosynthetic autotrophic bacteria oxidise nitrates, nitrites and ammonia and use the released energy for their ATP production.
c. Cyanobacteria often form blooms in polluted water bodies.
d. Thermoacidophiles are found in extreme salty areas and cannot survive in water having low pH.

In the light of above statements, select the **correct** option.

(1) All are correct except d (2) Only a and b are correct
(3) Only c and d are correct (4) Only b and c are correct

171. Match column I and column II w.r.t various theories of origin of life.

	Column I		**Column II**
a.	Theory of special creation	(i)	Disapproved by Louis Pasteur
b.	Theory of panspermia	(ii)	Diversity was always the same since creation and will be the same in future
c.	Theory of chemical evolution	(iii)	Life came from pre-existing non-living organic molecules
d.	Theory of spontaneous generation	(iv)	Units of life called spores were transferred from different planets to the Earth

Choose the **correct** option.

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iv), b(iii), c(i), d(ii) (4) a(ii), b(iii), c(iv), d(i)

172. Read the following statements

I. Morphine is obtained from *Erythroxylum coca* which is native of South America.

II. Morphine has a potent stimulating action on the CNS, producing a sense of euphoria and increased energy.

Select the correct option.

(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statements are correct
(4) Both statements are incorrect

173. Choose the **correct** statement.

(1) Proto oncogenes are identified mainly in cancer cells, which lead to oncogenic transformation.
(2) Viral DNA of HIV incorporates into genome of macrophages which leads to production of new viral RNA
(3) Enzyme linked immuno-sorbent assay is an antigen-antibody test, used only for the diagnosis of Acquired Immuno Deficiency Syndrome.
(4) Techniques of molecular biology are still not advanced enough to detect genes in individuals with inherited susceptibility to certain cancers.

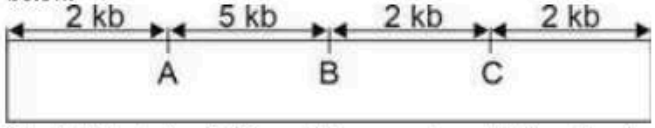
174. Read the following and select the correct option.

(a) Macrophages are tissue-resident cells
(b) Natural passive immunity is observed when IgG is passed from mother foetus during pregnancy.
(c) Colostrum is enriched in the dimeric immunoglobulin.

(1) (a), (b) and (c) are correct
(2) Only (a) and (b) are correct
(3) (a) and (c) are incorrect
(4) Only (b) and (c) are correct

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175. Consider the segment of the DNA shown in the figure given below.



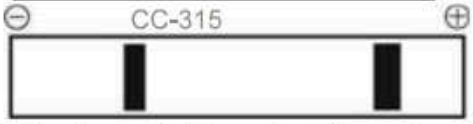
The labelled sites A, B and C represent restriction sites for *Pvu I*, *Sma I* and *Pvu II* respectively.

Now, if we perform complete restriction digestion of the given segment with only *Sma I* and *Pvu II*, then, which of the following will correctly represent the results obtained after performing gel electrophoresis?

(1)


(2)


(3)


(4)


176. Select the incorrect statement w.r.t bioreactors.

(1) In a continuous culture system, used medium is drained out from one side while fresh medium is added from the other side
(2) The most commonly used bioreactors are of stirring type.
(3) A stirred-tank reactor is always rectangular in shape to facilitate the mixing of the reactor's contents.
(4) Aeration and agitation ensures optimum oxygen levels for aerobic fermentation.

177. Choose the incorrect match.

(1) *Hind* II - Recognises a specific sequence of four base pairs
(2) Transformation - Procedure through which a piece of alien DNA is introduced in a host bacterium
(3) *Eco*RI – Produces staggered ends
(4) Bioprocess engineering - Maintenance of sterile ambience

178. A certain circular plasmid had two restriction sites, one each for a different enzyme.

This plasmid was tried to get digested by using a restriction enzyme which did not have a recognition site in the plasmid.

Which of the following would be present in the mixture after the restriction digestion?

(1) Two DNA fragments
(2) Single linear fragment of DNA
(3) Undigested plasmid
(4) Three circular DNA fragments

179. Consider the following features w.r.t. insulin pro-hormone.

(a) 'A' peptide
(b) Disulphide bonds
(c) 'B' peptide
(d) 'C' peptide

How many of the above structures does it lack?

(1) One
(2) Two
(3) Three
(4) Zero

180. Select the **correct** statement.

(1) RNAi is a post transcriptional gene silencing method
(2) Transgenic tomatoes have been developed to reduce pre-harvest losses by decreasing their shelf lives
(3) In GMO, early exhaustion of fertility of soil is promoted
(4) Gene *crylAb* controls infection by *Meloidogyne incognita* in tobacco plants

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